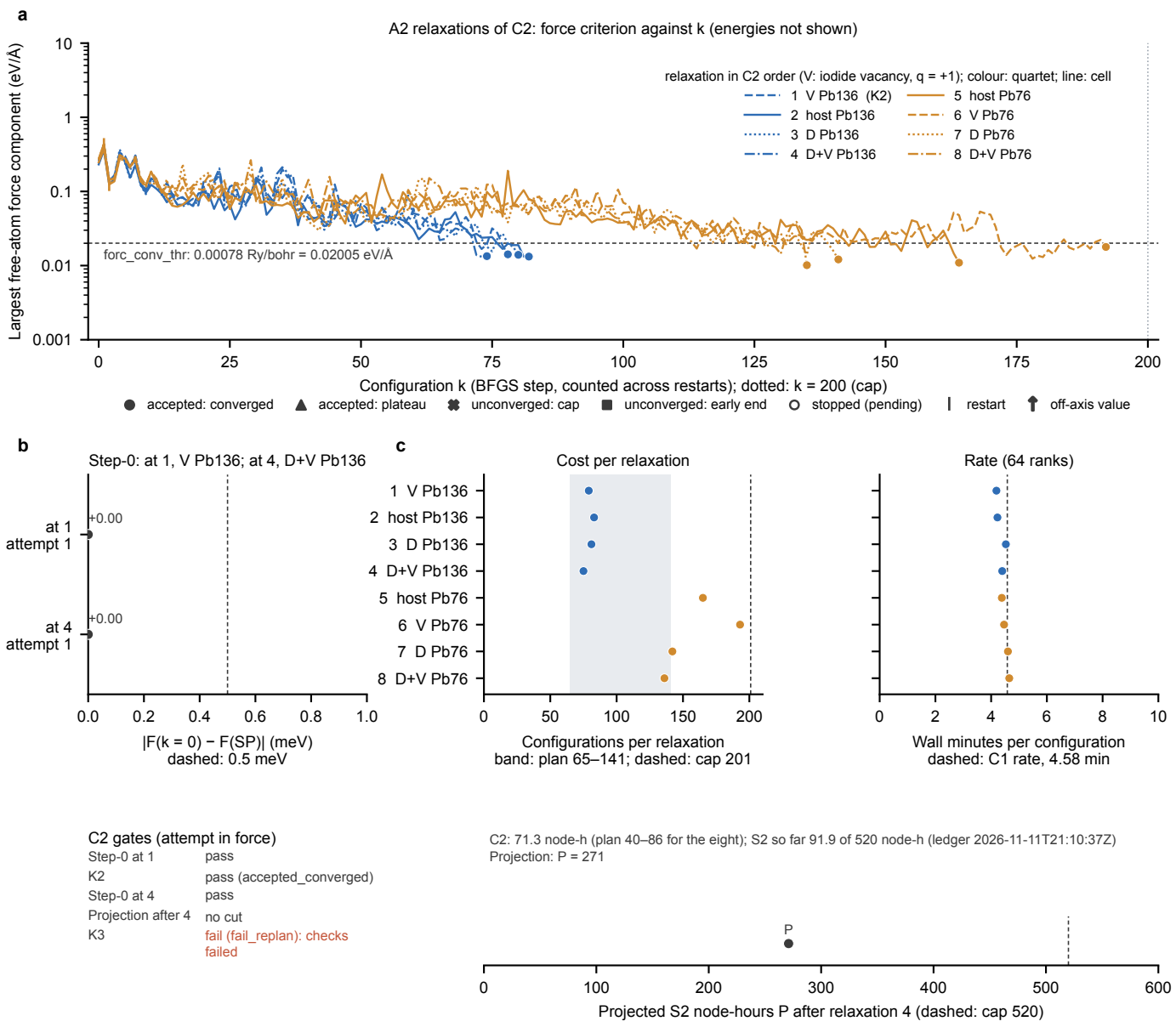

a
A2 relaxations of C2: force criterion against k (energies not shown)
Largest free-atom force component (eV/Å)
relaxation in C2 order (V: iodide vacancy, q = +1); colour: quartet; line: cell
1 V Pb136 (K2)
2 host Pb136
3 D Pb136
4 D+V Pb136
5 host Pb76
6 V Pb76
7 D Pb76
8 D+V Pb76
forc_conv_thr: 0.00078 Ry/bohr = 0.02005 eV/Å
Configuration k (BFGS step, counted across restarts); dotted: k = 200 (cap)
accepted: converged
accepted: plateau
unconverged: cap
unconverged: early end
stopped (pending)
restart
off-axis value
b
Step-0: at 1, V Pb136; at 4, D+V Pb136
at 1 attempt 1
at 4 attempt 1
+0.00
+0.00
|F(k = 0) − F(SP)| (meV)
dashed: 0.5 meV
c
Cost per relaxation
Configurations per relaxation
band: plan 65–141; dashed: cap 201
Rate (64 ranks)
Wall minutes per configuration
dashed: C1 rate, 4.58 min
C2 gates (attempt in force)
Step-0 at 1 pass
K2 pass (accepted_converged)
Step-0 at 4 pass
Projection after 4 no cut
K3 fail (fail_replan): checks failed
C2: 71.3 node-h (plan 40–86 for the eight); S2 so far 91.9 of 520 node-h (ledger 2026-11-11T21:10:37Z)
Projection: P = 271
P
Projected S2 node-hours P after relaxation 4 (dashed: cap 520)